

Thm. Let $\{a_n\}$ be a series, and take $R = (\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}})^{-1}$.

Then, the series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely if $|x| < R$,
diverges if $|x| > R$.

Therefore, we can define $f: (-R, R) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n x^n$.

For any $0 < \epsilon < R$, f converges uniformly on $[-R + \epsilon, R - \epsilon]$ and
 f is differentiable on $(-R, R)$, with

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

proof. The first assertion is given by the root test.

And, the second assertion is given by the Weierstrass-M test, specifically,
for all $n \in \mathbb{N}$ and $x \in [-R + \epsilon, R - \epsilon]$, $|a_n x^n| = |a_n| |x|^n \leq |a_n| \cdot |R - \epsilon|^n$, and

$$\sum_{n=0}^{\infty} |a_n| \cdot |R - \epsilon|^n$$

converges, by the first. To show the third assertion, observe that

$$\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \limsup_{n \rightarrow \infty} |n a_n|^{\frac{1}{n}}$$

Because: since $\lambda_n = n^{\frac{1}{n}} - 1$ satisfies $(1 + \lambda_n)^n = n \geq \frac{n(n-1)}{2} \cdot \lambda_n^2$, thus

$$\lambda_n^2 \leq \frac{2}{n-1} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Thus $n^{\frac{1}{n}} \rightarrow 1$ as $n \rightarrow \infty$. Given $\epsilon > 0$, there is $N_1 \in \mathbb{N}$ s.t.

$$n \geq N_1 \Rightarrow |a_n|^{\frac{1}{n}} < \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} + \frac{\epsilon}{2}$$

Meanwhile, since $n^{\frac{1}{n}} \rightarrow 1$ as $n \rightarrow \infty$, take $N_2 \in \mathbb{N}$ s.t.

$$n \geq N_2 \Rightarrow n^{\frac{1}{n}} < 1 + r, \text{ where } r = \frac{A + \epsilon}{A + \frac{\epsilon}{2}} - 1, \quad A = \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}.$$

Then,
 $n \geq \max\{N_1, N_2\} \Rightarrow n^{\frac{1}{n}} \cdot |a_n|^{\frac{1}{n}} \leq \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} + \epsilon.$

Moreover, there are infinitely many $n \in \mathbb{N}$ s.t. $\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} - \frac{\epsilon}{2} < |a_n|^{\frac{1}{n}}$,

we can choose similarly, 'infinitely many $n \in \mathbb{N}$ s.t. $1 - t < n^{\frac{1}{n}}$, $t = 1 - \frac{A - \epsilon}{A - \frac{\epsilon}{2}}$

Now proved. Using this, $\sum n a_n x^{n-1}$ converges uniformly in $(-R, R)$,

we obtain the result.

Thm. Suppose that $f: (-R, R) \rightarrow \mathbb{R} : x \mapsto \sum_{n=0}^{\infty} a_n \cdot x^n$ is well-defined.

Given $c \in (-R, R)$, the equality holds:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} \cdot (x-c)^n \quad (x \in B_{R-|c|}(c)).$$

Proof. Observe that

$$f(x) = \sum_{n=0}^{\infty} a_n \cdot [(x-c) + c]^n = \sum_{n=0}^{\infty} a_n \cdot \sum_{m=0}^n \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m.$$

And, since $x \in B_{R-|c|}(c) \Leftrightarrow |x-c| + |c| < R$,

$$\sum_{n=0}^{\infty} \sum_{m=0}^n |a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m| = \sum_{n=0}^{\infty} |a_n| \cdot [|x-c| + |c|]^n < \infty.$$

So, $\sum_{m=0}^{\infty} |A_{nm}|$ converges for each n , where $A_{nm} = a_n \cdot \binom{n}{m} \cdot c^{n-m} \cdot (x-c)^m$.

We can change the limits

$$f(x) = \sum_{m=0}^{\infty} \left(\sum_{n=m}^{\infty} \binom{n}{m} a_n \cdot c^{n-m} \right) (x-c)^m = \sum_{m=0}^{\infty} \frac{f^{(m)}(c)}{m!} \cdot (x-c)^m.$$

Thm. Given series $\sum a_n$ and $\sum b_n$, define $C_n = \sum_{k=0}^n a_k \cdot b_{n-k}$.

If $\sum_{n=0}^{\infty} a_n$ Converges absolutely and $\sum_{n=0}^{\infty} b_n$ Converges, then

$$\sum_{n=0}^{\infty} C_n = \left(\sum_{n=0}^{\infty} a_n \right) \cdot \left(\sum_{n=0}^{\infty} b_n \right).$$

Proof. Denote

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k, \quad B = \sum_{n=0}^{\infty} b_n \quad \text{and} \quad \beta_n = B_n - B$$

Then,

$$\begin{aligned} \sum_{k=0}^n a_k &= a_0 b_0 + (a_0 b_1 + a_1 b_0) + \dots + (a_0 b_n + a_1 b_{n-1} + \dots + a_n b_0) \\ &= a_0 B_n + a_1 B_{n-1} + \dots + a_n B_0 \\ &= a_0 (B + \beta_n) + a_1 (B + \beta_{n-1}) + \dots + a_n (B + \beta_0) \\ &= A_n B + a_0 \beta_n + a_1 \beta_{n-1} + \dots + a_n \beta_0. \end{aligned}$$

Now, enough to show that

$$\gamma_n = a_0 \beta_n + a_1 \beta_{n-1} + \dots + a_n \beta_0$$

Converges to zero.

Let $\epsilon > 0$ be given. Since $\beta_n \rightarrow 0$ as $n \rightarrow \infty$, there exists $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow |\beta_n| \leq \epsilon$.

Now, for $n \geq N$,

$$\begin{aligned} |\gamma_n| &\leq |\beta_0 a_n + \dots + \beta_N a_{n-N}| + |\beta_{N+1} a_{n-N+1} + \dots + \beta_n a_0| \\ &\leq |\beta_0 a_n + \dots + \beta_N a_{n-N}| + \epsilon \cdot (|a_{N+1}| + \dots + |a_0|) \\ &\leq |\beta_0 a_n + \dots + \beta_N a_{n-N}| + \epsilon \cdot \sum_{k=0}^{\infty} |a_k| \quad \leftarrow \end{aligned}$$

Using this, Converges absolutely.

Take $\limsup$, then

$$\begin{aligned} \limsup_{n \rightarrow \infty} |\gamma_n| &\leq \limsup_{n \rightarrow \infty} |\beta_0 a_n + \dots + \beta_N a_{n-N}| + \epsilon \sum_{k=0}^{\infty} |a_k| \\ &= \epsilon \cdot \sum_{k=0}^{\infty} |a_k|. \end{aligned}$$

Consequently, $\gamma_n \rightarrow 0$ as $n \rightarrow \infty$, so that we obtain the result.

Abel.

Thm. Let $f: [-1, 1) \rightarrow \mathbb{R}; x \mapsto \sum_{n=0}^{\infty} a_n x^n$ be defined.

If $\sum_{n=0}^{\infty} a_n$ converges, then $\lim_{x \rightarrow 1^-} f(x) = \sum_{n=0}^{\infty} a_n$.

Proof. Put $A_n = \sum_{k=0}^n a_k$, and $A = \lim_{n \rightarrow \infty} A_n = \sum_{n=0}^{\infty} a_n$. As a Cauchy product,

$$\frac{1}{1-x} f(x) = \sum_{n=0}^{\infty} A_n \cdot x^n, \text{ so } f(x) = (1-x) \cdot \sum_{n=0}^{\infty} A_n x^n \quad (-1 < x < 1)$$

and since $(1-x) \cdot \sum_{n=0}^{\infty} x^n = 1$, so

$$\begin{aligned} f(x) - A &= (1-x) \sum_{n=0}^{\infty} A_n x^n - A \cdot (1-x) \sum_{n=0}^{\infty} x^n \\ &= (1-x) \left(\sum_{n=0}^{\infty} A_n x^n - A \cdot \sum_{n=0}^{\infty} x^n \right) \\ &= (1-x) \cdot \left(\sum_{n=0}^{\infty} (A_n - A) \cdot x^n \right), \quad (-1 < x < 1). \end{aligned}$$

Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow |A_n - A| < \varepsilon$, and put $M = \max \{|A_n - A| \mid n = 0, \dots, N\}$.

$$\begin{aligned} \text{Now, } |f(x) - A| &\leq \left| (1-x) \cdot \sum_{n=0}^N (A_n - A) x^n \right| + \left| (1-x) \cdot \sum_{n=N}^{\infty} (A_n - A) x^n \right| \\ &\leq (1-x) \cdot MN + \varepsilon \end{aligned}$$

Therefore, $\lim_{x \rightarrow 1^-} f(x) = A$.

이제, 정리도 지루했던 논의는 종결이다.

[Thm. Suppose that $\sum a_n x^n$ and $\sum b_n x^n$ converge in $(-R, R)$.

Let $E \subseteq (-R, R)$ be a set such that

$$x \in E \iff \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} b_n x^n.$$

If E has a limit point in $(-R, R)$, then $a_n = b_n$ for all $n \geq 0$.

Proof. Put $C_n = a_n - b_n$, and let $f(x) = \sum_{n=0}^{\infty} C_n x^n$ ($x \in S$)

Then, $f(x) = 0$ on E . Let $A = E' \cap (-R, R)$, and $B = (-R, R) \setminus A$.

Since E' is closed, $B = (-R, R) \cap E'^c$ is open. And, if A is open, then

$\{A, B\}$ is separation of an interval $(-R, R)$ so A or B must be empty.

By the hypothesis, $A \neq \emptyset$, so, $B = \emptyset$, $A = E' \cap (-R, R) = (-R, R)$.

Now, since f is continuous in $(-R, R)$ so